

CSC5041: Assignment 1
Deadline: October 28, 2026
Total marks: 15

Question 1 (5 marks)

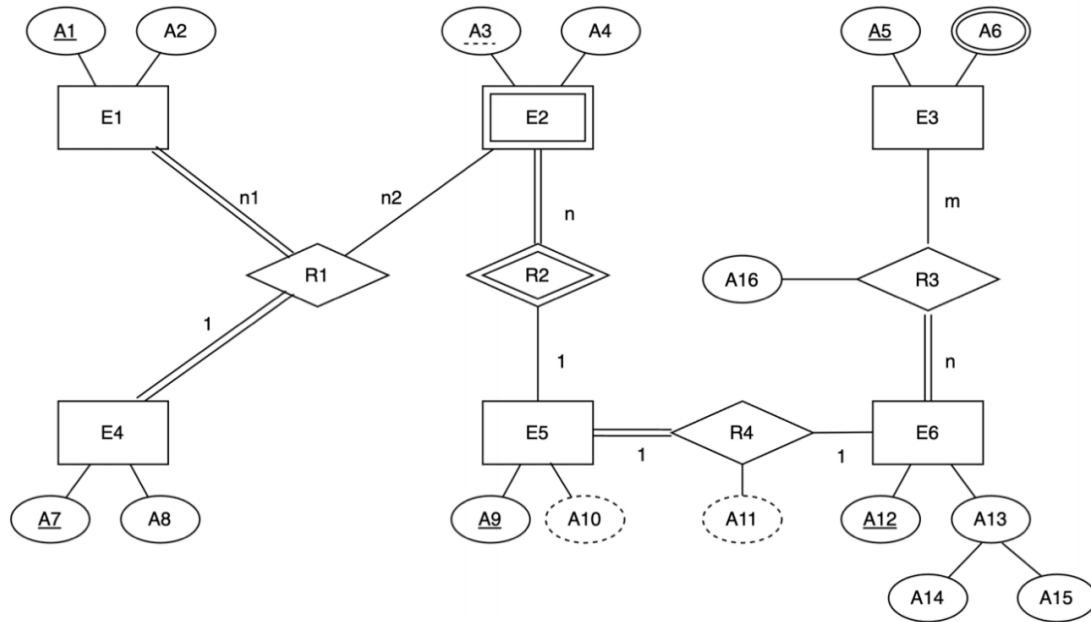
A hospital outpatient booking platform seeks your assistance in building a modern database. They provide a set of requirements as follows:

- Each **hospital** is identified by a unique hospital code, and stores its name and location, which includes district and city.
- Every hospital must have one or more **departments**, and each department is identified by a department number unique to its hospital. A department also stores its floor and its type (e.g., outpatient, inpatient).
- Each **doctor** is uniquely identified by a doctor ID. A doctor stores their name, title and licence number. Each department employs one or more doctors, and each doctor works in exactly one department.
- A **consultation slot** is identified by a combination of slot number and slot date, and stores whether it is a morning or an afternoon session. Each consultation slot is held in exactly one department, and a department holds multiple consultation slots. The recorded start time and end time of the slot are stored, and the duration of the slot is also recorded.
- A consultation slot contains one or more **queue positions**, and each queue position is uniquely identified by its position number within the slot. Each queue position records its status (e.g., available, assigned, closed) and the estimated waiting time in minutes, and may carry one or more tags (such as priority, follow-up, walk-in, paediatric, or isolation).
- A **patient** is identified by a patient ID and stores personal information including full name with first name and last name, gender, date of birth, age, email address, phone number, and address with district, city and postcode.
- An **appointment** is uniquely identified by an appointment ID. Each appointment is made by a single patient but may include multiple consultation slots, such as a follow-up series or a multi-department referral. A consultation slot can appear in multiple appointments. For each consultation slot included in an appointment, the patient is assigned one queue position. Each queue position of a consultation slot can be assigned at most once; however, some queue positions may remain unassigned. An appointment also stores the booking timestamp, the total fee, the payment method, and the total number of slots included. A patient may have zero or more appointments.

Your task: draw an ER diagram to represent the scenario, clearly state any *reasonable assumptions* that you choose to make. Please keep to the notations taught in the lecture.

Question 2 (4 marks)

Convert the ER-diagram below into a relational data model. Use the notations/model taught in the lecture.



Question 3 (6 marks)

Consider the following schema for a ride-hailing and bike-sharing system:

- **Rider**(riderID, riderName, membershipTier, age)
- **VehicleType**(typeID, typeName, category, baseFare)
- **Zone**(zoneID, zoneName, city)
- **Station**(stationID, zoneID, capacity, stationType)
- **Trip**(tripID, riderID, typeID, startStationID, endStationID, tripDate, distance, fare)
- **Payment**(paymentID, riderID, payDate, amount)

Your task: write the relational algebra expressions for the following queries:

- 1) Find the names of riders who made car trips during June 2025. (1 mark)
- 2) Find the maximum payment amount of each rider who has made only trips with a fare ≤ 10 . (1 mark)
- 3) Find the membership tier held by each rider who has taken trips using every vehicle category available in the system (e.g., car, bike, scooter). Assume that each rider holds exactly one membership tier (e.g., adult, student, senior). (2 marks)
- 4) Find the average age of riders (allowing repeated trips) who started from any station located in zone “Xuhui” during the period from 2025-10-01 to 2025-11-02 (inclusive). That is, if a rider started multiple times in this period, their age is counted multiple times in the average. (2 marks)

Note: we only allow the operators used in the lecture slides.

Note: attributes that are not part of the primary key are not unique. To elaborate, attributes such as the name of the riders (**riderName**), the name of the stations (**stationType**), or the name of the vehicle types (**typeName**) should not be treated as unique identifiers.

Assignment Submission

- You must submit an electronic copy of your answers to the above questions to the course website.
- We only accept .pdf files.
- Please name your files in the following format: 226010XXX-Assg1.pdf, replacing 226010XXX with your own student ID.

Note:

- 1) If you have problems relating to your submission, please contact the teaching assistant on WeChat or Email.
- 2) All submissions will be checked for plagiarism. The university regards plagiarism as a form of academic misconduct and has very strict rules. Not knowing the rules will not be considered a valid excuse when you are caught.

Late Submission Penalty

- A hefty 5% of your final mark will be deducted for each additional day (24hr) after the specified submission time and date.
- Submissions that are more than three days late will not be marked.